



MRS6240-P2512

mmWave Radar Module

User Guide

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1 Overview

1.1 Module Introduction

MRS6240-P2512 is a module developed based on the new generation 60GHz 2T4R FMCW AIP radar sensor chip RS6240 from Possumic. The module integrates a 2T4R AiP radar antenna array, crystal oscillator, and flash, enabling developers to quickly develop and apply the module. The module includes four low-power management modes: shutdown, standby, idle, and working. It not only features low power consumption in standby mode but also benefits from a fast burst engine and high-performance RF transceiver design, achieving low power consumption during sensing. The flexible configuration of these modes allows the entire module's power consumption to meet various demanding application requirements. Its configuration and data acquisition are implemented through UART/SPI/I2C digital interfaces. The module is divided into three subsystems:

- mmWave RF and BB Subsystem: This module encompasses all high-performance radar RF/analog and base band digital signal circuits. It also incorporates a linear frequency modulated pulse generator and a Tx-to-Rx state machine.
- Processing and Application Subsystem: Consists of a 32-bit RISC CPU and a radar hardware accelerator (HWA) for various application processing and task management. The proprietary HWA IP is used to handle specific radar computations (such as FFT, CFAR, etc.), compression, as well as moving target detection and location calculation.
- Power and clock subsystem: RS6240 is a complex single-chip SoC with multiple functional subsystems, which are centrally managed internally through an integrated PMU system and clock subsystem.

1.2 Module Specifications

Table 1-1 Module Specifications

Specifications		Minimum	Typical	Maximum	Unit	Remarks
Radar Specifications	Operating Frequency Band	57	-	66	GHz	-
	Continuous Frequency Scanning Bandwidth	-	8	-	GHz	-
	Number of transmitting and receiving channels	2T 4R			-	-
	Equivalent Isotropic Radiation Power (EIRP)	-	17	-	dBm	
	Horizontal FOV	-	120	-	°	
	Vertical FOV	-	120	-	°	
	The Farthest Detection Distance	-	-	20	m	It depends on the size of the RCS of the detection target
	Recent detection distance	-	-	0.01	m	Take human targets as a reference
Hardware	Module Dimensions	25x12			mm ²	-

Specifications	Power Supply Voltage	1.7	3.3	5.5	V	-
	Operating Temperature	-40	-	125	°C	Operating junction temperature

1.3 Application Scenarios

The MRS6240 module can be widely applied in various AIOT intelligent perception scenarios, such as smart home appliances, intelligent security, smart health care, intelligent cockpits, etc., to achieve high-precision and high-sensitivity detection of human body and object movements, micro-motions, and so on.

- Human movement/Presence detection
- Human body tracking and counting
- Video doorbell/Network camera
- Air conditioner
- Obstacle avoidance for drones/Robot
- Detection of children left inside vehicles

2 System Application Block Diagram

The MRS6240 millimeter-wave radar module comprises a power supply and eight GPIO as peripherals. Specifically, VCC serves as the main power supply for the module, while VDDIO provides power for the GPIO and peripherals on the module. When the power supply voltage does not exceed 3.6V, VCC and VDDIO can be directly short-circuited during application.

V13 can be powered by either the internal VINT or an external low-power DCDC.

The MRS6240 millimeter-wave radar module integrates a CPU, enabling it for application development. Therefore, the configuration can choose to include or exclude an external MCU.

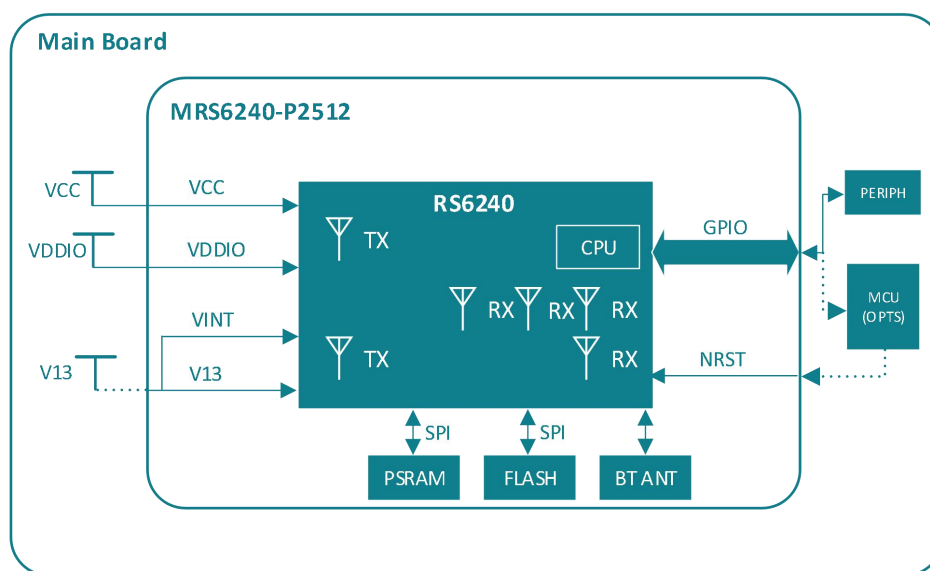


Figure 2-1 Application Block Diagram of MRS6240-P2510 System

3 Pin Definition

3.1 Pin Definition

The layout of the module's pin is shown in the figure below.

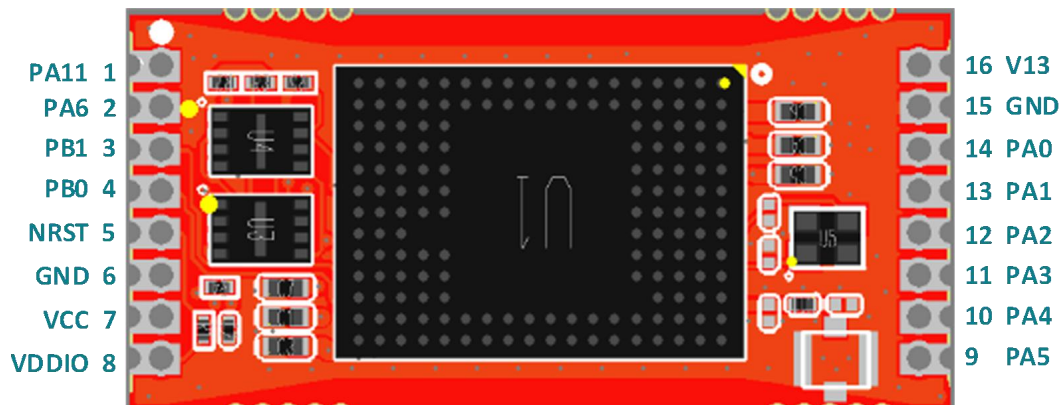


Figure 3-1 Pin Layout

3.2 Pin Description

The module has 16 pins, and the corresponding function descriptions are shown in the table below.

Table 3-1 Description of module pin functions

Pin Number	Name	Type	Function
1	PA11	IO/WUP	PA11,WIO11,I2C1_SCL,CANFD_RX,PWM3,GPADC3
2	PA6	IO/WUP	PA6,WIO6,UART1_RTS,UART2_TX,I2C1_SDA,PWM6,GPADC6
3	PB1	IO/WUP	PB1,WIO17,UART0_RX,PWM1
4	PB0	IO/WUP	PB0,WIO16,UART0_TX,PWM0
5	NRST	I	Reset signal of radar module
6	GND	G	Ground Pin, Short directly to module ground plane
7	VCC	PI	Module main power supply, 1.7V~5.5V
8	VDDIO	PI	VDDIO is the power supply for GPIO and peripherals on the module, ranging from 1.7V to 3.6V. When the power supply voltage does not exceed 3.6V, VCC and VDDIO can be directly short-circuited during application.
9	PA5	IO/WUP	PA5,WIO5,SPI0_WP,I2C0_SCL,PWM5
10	PA4	IO/WUP	PA4,WIO4,SPI0_HOLD,I2C0_SDA,PWM4
11	PA3	IO/WUP	PA3,WIO3,SPI0_MISO,UART1_CTS,I2C0_SCL,UART2_RX/LIN_RX,PWM3
12	PA2	IO/WUP	PA2,WIO2,SPI0_MOSI,UART1_RTS,I2C0_SDA,UART2_TX/LIN_TX,PWM2
13	PA1	IO/WUP	PA1,WIO1,SPI0_CLK,UART1_RX,CANFD_RX,PWM1
14	PA0	IO/WUP	PA0,WIO0,SPI0_CS,UART1_TX,CANFD_TX,PWM0
15	GND	G	Ground Pin, Short directly to module ground plane

16	V13	PI/PO	For the module V13 power supply, when using an external DCDC, it supplies 1.3V. When using an internal LDO, this network measures 1.5V.
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Note:

- **G** represents the ground
- **PI** stands for Input Power
- **PO** stands for output power
- **GPIO** stands for General-Purpose Digital Input/Output
- **I** represents the input pin
- **WUP** stands for a digital pin that can maintain a state or wake up the system in standby mode, where WIOx refers to the sequential number of the WUP type IO.

4 Peripheral

4.1 GPIO

The MRS6240-P2512 module provides 10 GPIO, each with multiple reusable functions.

- Three modes: input, output, and high impedance
- Input configurable pull-up and pull-down
- The output drive capability is set to 20mA by default, with one setting at 20mA and a maximum configuration of 80mA
- The IO output supports both push-pull and open-drain modes

Note:

- In low-power scenarios, all GPIO support triggering the system wake-up through interrupt mode.

4.2 UART

The MRS6240-P2512 module supports UART0 and UART1. UART is compatible with the 16550 industry standard and supports baud rates up to 3Mbps.

- UART has the following characteristics:
- Programmable transmit and receive baud rate
- Baud rate self-checking function
- Supports 64-byte receive and transmit FIFO
- Supports data lengths of 5/6/7/8 bits
- Support 1/1.5/2 stop bits
- Support parity check
- Supports DMA high-speed data communication
- Supports UART wake-up function, UART RX has the ability to detect rising edge changes and trigger interrupts, with the number of rising edges detected configurable from 1 to 8. In low-power scenarios, the system is woken up after UART RX receives a matching rising edge.

4.3 I2C

The MRS6240-P2512 module supports I2C0 interface communication. I2C has the following characteristics:

- Can be configured as master and slave modes
- Supports standard mode (100Kbit/s)
- Supports high-speed mode (400Kbit/s)
- Support high-speed mode enhancement (1000Kbit/s)
- Supports 7-bit addressing mode and 10-bit addressing mode
- Supports receive and transmit FIFO depths of 8 bytes
- Supports I2C wake-up function. In low-power scenarios, I2C supports 7-bit addressing matching to trigger the wake-up system.

4.4 PWM

The PWM controller can be used to output modulated digital waveform signals, supporting an adjustable duty cycle from 0% to 100%.

4.5 SPI

RS6240 supports a quad-serial SPI interface (QSPI). The QSPI protocol supports half-duplex, full-duplex, synchronous, and serial communication with peripherals. This interface can be configured in either master or slave mode. There are four possible combinations of serial clock phase and polarity. The following features are supported:

- The clock frequency can reach up to 96 MHz
- Support Motorola SPI mode
- Programmable master mode or slave mode
- Supports standard SPI, dual SPI, and quad SPI modes
- Programmable data width, ranging from 4 to 32 bits
- The FIFO depth for both sending and receiving data is 32 words
- Programmable sampling delay time in main mode
- Programmable Serial Clock Polarity (SCPOL) and Serial Clock Phase (SCPH)
- Supports interrupt and DMA
- Programmable instructions, address length, and wait cycles in main mode
- Support clock detection to wake up the system from standby state

5 Hardware Usage

5.1 Hardware Start Up

Module VCC supplies 3.3V, VDDIO supplies 3.3V, and V13 is internally powered.

NRST is first pulled low, then released and pulled high, and the hardware is powered on to boot. The firmware is loaded from the external Flash and executed in SRAM.

5.2 Firmware Burning

Based on the specific hardware configuration, users can enable the chip to enter the Burning mode through the following two schemes.

Option 1: The main controller can control the NRST pin through IO

- 1) Keep VCC powered on, with PB0 and PB2 at high voltage levels;
- 2) Set NRST to low level and delay for 20ms;
- 3) Set NRST to high level and delay for 50ms;
- 4) Set PB0 (or PB2) to low level, and set NRST to low level;
- 5) Delay for 20ms, set NRST to high level;
- 6) Delay for 50ms, set PB0 (or PB2) to high level. At this point, the chip and flash have completed resetting and entered Burning mode.

Option 2: Suitable for systems with an external control power supply interruption mechanism

- 1) Power off our chips, Flash, or modules. If the external capacitor is 10uF, it is recommended to maintain the power-off state for 10 seconds. Please adjust the power-off duration based on the external capacitor and the actual power-down speed;
- 2) Set PB0 (or PB2) to low level;
- 3) Power on our chip and Flash or module again, delay for 50ms, and set PB0 (or PB2) to high level. At this point, the chip and flash have completed resetting and entered Burning mode.

Finally, through the BurnTool, the firmware is written into the FLASH using any of the interfaces: UART (PB0/PB1), SPI (PA0~PA3), and I2C (PA4/PA5).

6 Electrical Characteristics

6.1 Power Supply Operating Conditions

Table 6-1 Operating Conditions of Module Power Supply

Name	Parameter Description	Minimum	Typical Value	Maximum	Unit
VCC	Module Power Supply Input Voltage	1.7	3.3	5.5	V
VDDIO	Module IO Input Voltage	1.7	3.3	3.6	V
V13	Module Internal Power Supply Output Voltage	1.2	1.5	2.5	V
Ivcc	The Current Provided By the External Power Supply	-	-	800	mA

7 Module Size and PCB Packaging

7.1 Module Dimensions

Unit:mm

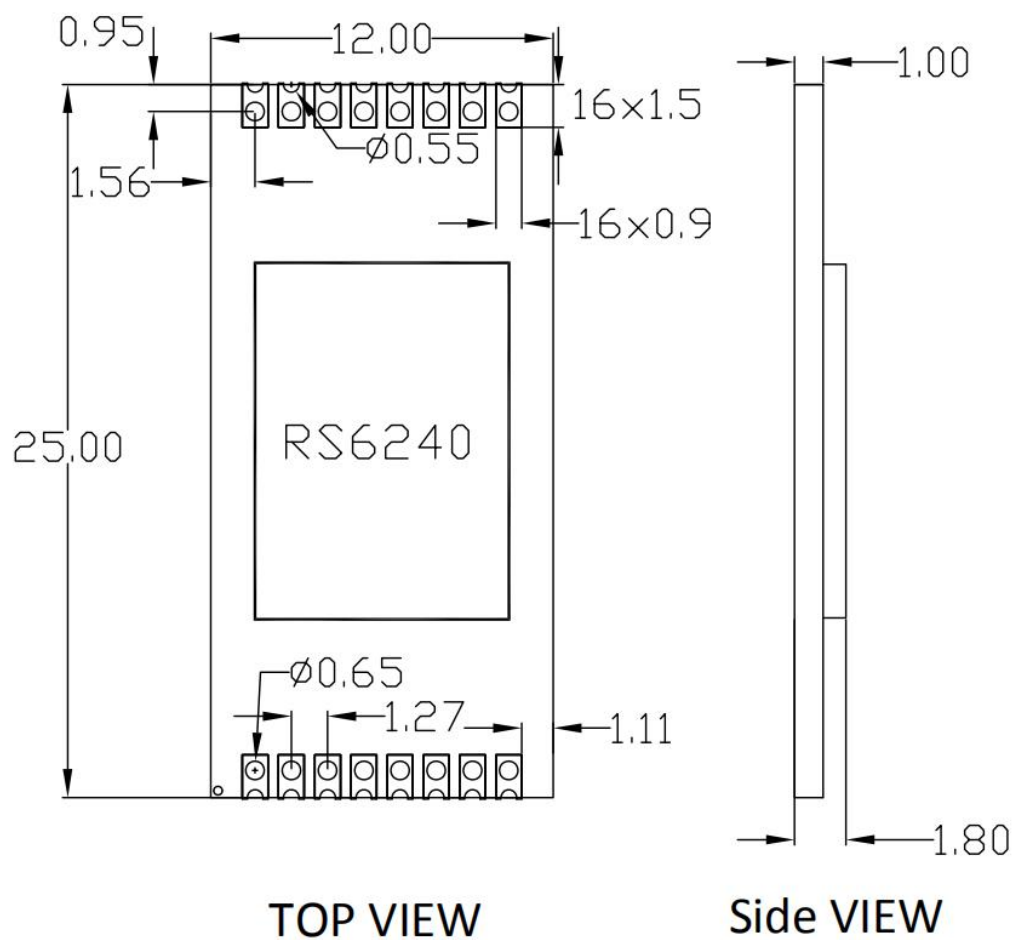


Figure 7-1 Dimensions of MRS6240-P2510 Module

7.2 Recommended PCB Packaging

Unit:mm

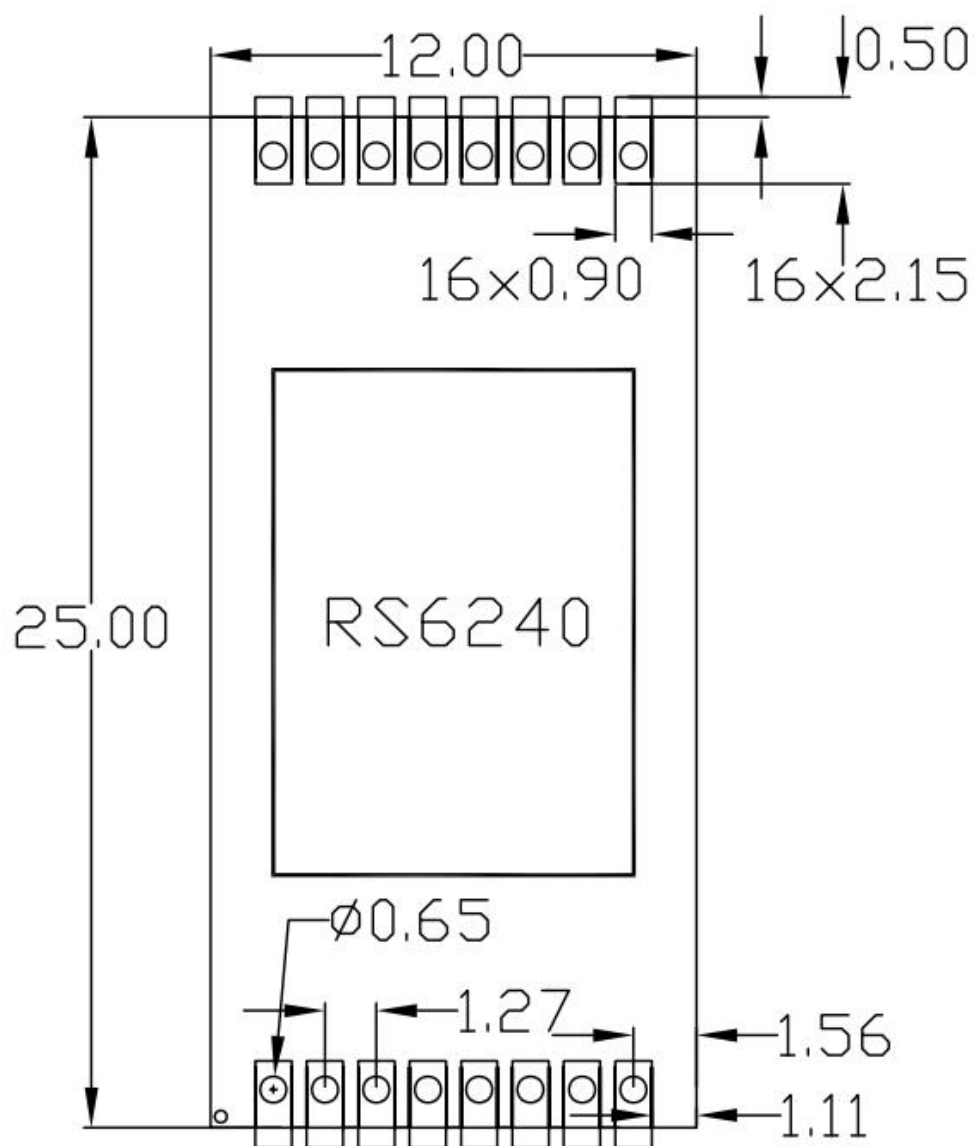


Figure 7-2 Recommended PCB Packaging

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